MC 451 Codebook and Qualitative Memo: Worked Example

- Rebuilt from my own research project, Play to Learn in VR (P01.01), continuing the student study from the Project Prospectus and Definitions Practice examples: one coder reading 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit: the codebook and the memo as one document with two headed parts. Every example tweet is paraphrased from a real tweet in the archive; none is quoted, so a search cannot lead back to the account.
- The rubric map at the end shows which grading row each part earns and why.

The example

Part 1: Codebook

Study overview

This codebook measures whether tweets about playing in virtual reality (VR) that refer to learning are more often positive toward VR than tweets that do not, and which benefits and obstacles of VR those learning tweets name. The material is English-language tweets, not retweets, mentioning VR with play, from the Play to Learn in VR archive (Twitter Academic Research API). The time period is calendar year 2021.

Unit of analysis

One tweet: the text the author wrote in that single post. Quoted tweets, links, images, and replies are not part of it. Every variable below is coded on one tweet. Three fields are looked up, not coded: the tweet ID, the posting date, and the like count.

Variable 1: Learning reference

Element	Learning reference
Conceptual definition	Whether the tweet is about someone gaining or passing on a skill or knowledge in or through VR.
Operational definition	Read the whole tweet. Mark 1 if any clause describes a person learning, teaching, practicing, or being trained in something in or through VR; otherwise mark 0. Check what a learning word means in its sentence before marking.
Level of measurement	Nominal: two named categories with no order between them.
Coded values	0 = no learning reference. 1 = learning reference.

Element	Learning reference
Edge-case rules	1. A learning word inside a title or an idiom (learned my lesson), or "learn more" in a promotion, is coded 0 unless another clause describes learning. 2. Learning to play a VR game, or relearning it on new controllers, is coded 1, because the skill is gained in VR.

Variable 2: Tone toward VR

Element	Tone toward VR
Conceptual definition	The writer's evaluation of VR or of the VR experience the tweet describes: favorable, unfavorable, or neither.
Operational definition	Find every word or emoji that evaluates VR or a VR experience, including a game played in VR. Mark 3 if all are favorable, 1 if all are unfavorable, and 2 for both or none.
Level of measurement	Ordinal: three ordered categories, with no claim the steps are equal.
Coded values	1 = negative. 2 = neutral or mixed. 3 = positive.
Edge-case rules	1. Praise for a game played in VR counts as an evaluation of a VR experience. Praise for a game played outside VR does not. 2. An advertisement with no evaluative word is coded 2; one calling the experience amazing is 3, because the variable records what the tweet says.

Variable 3: Affordance named

Element	Affordance named
Conceptual definition	The benefit the tweet attributes to VR: something VR lets a person do, or do better.
Operational definition	Look for a clause that says what VR lets someone do or makes better, and match it to one category below. If two categories apply, mark the one with more of the tweet's words, or the first if equal. If no clause names a benefit, mark 0.
Level of measurement	Nominal: named categories with no order.
Coded values	0 = no benefit named. 1 = fun or engagement. 2 = hands-on interaction (moving, building, handling objects). 3 = doing it with others. 4 = practicing a skill. 5 = other benefit (exercise, wellbeing, anything else).
Edge-case rules	1. A feature named without saying what it lets a person do (multiplayer, hand tracking) is coded 0. 2. Learning to play a VR game better is coded 4, the same as practicing a clinical skill in VR, because the variable records the kind of benefit.

Variable 4: Constraint named

Element	Constraint named
Conceptual definition	The obstacle the tweet attributes to VR: something that stopped, limited, or discouraged someone from doing something in VR.
Operational definition	Look for a clause that names a problem with VR that limited or discouraged use, and match it to one category below. Break a tie between two categories as for variable 3. If no clause names a problem, mark 0.
Level of measurement	Nominal: named categories with no order.
Coded values	0 = no obstacle named. 1 = physical discomfort (motion sickness, eye strain, headache). 2 = hardware cost (cannot afford a headset or computer). 3 = technical performance (lag, crashes, setup trouble). 4 = other obstacle (space, hygiene, anything else).
Edge-case rules	1. Advice about a problem (building tolerance to motion sickness) is coded in that category even if the writer did not suffer it. 2. "I can't play, no headset yet" is coded 2 only when price is stated; otherwise it is coded 4.

Decision rules for the whole codebook

These rules settle the hard cases met in the first 30 tweets.

1. Before coding, check that the tweet is about virtual reality. If VR appears only inside another word, a username, or an unrelated abbreviation, mark it out of frame, code nothing, and replace it from the reserve list.

2. The archive strips digits, so a console name or a date can appear with its number missing. Code the words that remain and never fill in a missing word from a guess.

3. Code what the tweet says, not who sent it: an advertisement is coded by its words like any other tweet. A sexually explicit tweet is marked out of frame, replaced, and never used as an example.

Examples

Two examples per category, each paraphrased from a tweet in the archive.

Category	Examples (paraphrased)
Learning reference 1	(a) Strangers in a VR tag game taught a new headset owner to play. (b) A university says VR training lets students practice clinical skills.
Learning reference 0	(a) A promotion says to learn more about a VR shooting demo. (b) A film synopsis about a trainee trapped in a VR game.

Category	Examples (paraphrased)
Tone 3, positive	(a) A player calls a VR horror game great.
	(b) A gamer calls a VR title one of the most immersive they have tried.
Tone 2, neutral or mixed	(a) A student asks if a cabled headset reaches the hallway (no evaluation).
	(b) A game every VR owner must try strains the eyes (both).
Tone 1, negative	(a) Buggy motion controllers mean a player rarely uses VR.
	(b) A writer dislikes the idea of meeting in VR.
Affordance 0, none	(a) A player is buying a horror game again for VR.
	(b) A recap lists upcoming games, some in VR.
Affordance 1, fun or engagement	(a) A player calls a VR horror game spooky fun.
	(b) A player is excited to slice blocks in a rhythm game again.
Affordance 2, hands-on interaction	(a) A writer asks whether a VR driving game gives the feel of the road.
	(b) VR gets a social VR user standing instead of sitting.
Affordance 3, doing it with others	(a) A player wants more time in a social VR app with friends.
	(b) A group that cannot meet plans a tabletop game in VR.
Affordance 4, practicing a skill	(a) A VR app offers practice at public speaking.
	(b) A player tells a newcomer to practice a VR game in multiplayer first.
Affordance 5, other benefit	(a) An all-day rhythm game session counts as exercise.
	(b) VR helped a player through lockdown.
Constraint 0, none	(a) A player calls a VR horror game great.
	(b) A player calls a VR demo amazing.
Constraint 1, physical discomfort	(a) A new owner lies down, motion sick after VR tag.
	(b) A player advises newcomers to rest when sick and go back in until they adjust.
Constraint 2, hardware cost	(a) A player cannot afford a headset.
	(b) A user wants a better headset but says they are expensive.
Constraint 3, technical performance	(a) A player says the motion controllers always bug out.
	(b) A player's boundary setup resets almost every session.
Constraint 4, other obstacle	(a) A student's study room has no space for a cabled headset.
	(b) A child skipped a shared headset because of germs.

Single-coder note

I am the only coder, so every reliability claim I make is about my own consistency, not agreement between coders. I will recode 15 tweets, 10 percent of the 150, at least 48 hours later with my first codes hidden, and report percent agreement per variable.

I am using four safeguards. I wrote every rule before coding the sample. Each category has examples from the archive. I freeze the codebook before the full pass and log every later change with its date and reason. I code in tweet-ID order with the username and like count hidden.

With a second coder, I would double-code 30 tweets, 20 percent of the sample, and report Krippendorff's alpha for each variable. I would train the coder with 20 practice tweets from outside the sample, then settle each disagreement against the written rule. The finding most at risk is H1, because it rests on tone, the variable that needs the most judgment.

Part 2: Qualitative memo

Before writing the rules, I read 30 tweets drawn at random from the archive, 15 with a learning word and 15 without, and kept them out of the 150 the Sampling Plan will draw. The biggest surprise was how little the learning word means. Only five of the 15 learning-word tweets describe a person learning in VR, such as a player relearning a game on new controllers. The rest used the word for something else: a "learn more" promotion, a film synopsis with a trainee, machine-learning hashtags, and an adult-game advertisement that passed the archive's sensitive flag. That split is why learning reference is coded by reading, not by the word list.

Most tweets carried no evaluation of VR at all. The praise I did see went to one VR game, a demo or a horror title, rather than to VR itself, which is why tone counts a game played in VR as a VR experience; the few negatives were about setup trouble or about working in VR. Benefits were rare: fun, couch co-op with others, the feel of the controls, practicing on a crane simulator, and one player saying VR helped them through lockdown. Those became affordance categories 1 to 5. Obstacles were motion sickness, including advice on getting used to it, a boundary that kept resetting and a game too demanding for most computers, and a study room with no space for a cabled headset, which became constraint categories 1, 3, and 4. A follow-up search of the archive for price words found players who could not afford a headset, which became category 2.

Two of the 30 were not about VR at all, and digits had been stripped from the text. Each became a decision rule.

Rubric map

The codebook rows total 35 and the memo 15, graded together as one 50-point assignment.

Rubric row	Points	Where the example earns it	Why it earns the full row
Codebook: study overview	3	The study overview paragraph	Three sentences name all three things the row asks for: the question (learning tweets and tone, plus the benefits and obstacles named), the material (English tweets about playing in VR from a named archive and collection route), and the time period (calendar 2021).
Codebook: unit of analysis	8	The unit of analysis paragraph	One coded item is stated exactly, once: one tweet's own text, with what is not part of it named. Every variable is coded on that tweet, and the three looked-up fields describe the same tweet, so nothing is coded on an account or a thread.
Codebook: every variable	5	The four variable tables	Every variable carries all five elements: conceptual definition, operational definition, level of measurement, coded values, and edge-case rules, in that order, with at least two rules each.
Codebook: decision rules	7	The three decision rules	Each one is written as a rule a coder can apply ("mark it out of frame, code nothing, and replace it"), and each covers a hard call met in the material: off-topic matches, stripped digits, and advertisements and explicit text.
Codebook: examples	5	The examples table	Every category of all four variables has two examples, each paraphrased from a real tweet in the archive, none invented, and each clearly fits its category under the rules above it.
Codebook: single-coder note	7	The three single-coder paragraphs	Paragraph one says in plain words that I am the only coder and that every reliability claim is about my own consistency. Paragraph two names four safeguards, above the minimum of three, each from the rubric's list. Paragraph three gives the whole counterfactual: how many tweets, which statistic, how the coder is trained and disagreements settled, and which finding is most at risk and why.

Rubric row	Points	Where the example earns it	Why it earns the full row
Qualitative memo	15	The three memo paragraphs	About 290 words, inside the 200-to-300-word range, on patterns seen in 30 real tweets rather than the topic in general. Each codebook category traces back to something the memo describes seeing: the learning split to both learning reference values, the unevaluated tweets, the praise for VR games, and the complaints to the three tone values and tone rule 1, the five benefits to affordance categories 1 to 5, the four obstacles and the price search to constraint categories 1 to 4, and the off-topic and stripped-digit tweets to decision rules 1 and 2.

The codebook revises the Definitions Practice entries where the 30 tweets showed a gap: tone now counts a game played in VR as a VR experience, and learning reference gained the rule for relearning a game.